

Option A: Immunology and Disease

6. (a) In 2010, an earthquake hit the country of Haiti in the Caribbean, causing devastation and severe damage to the water supplies and sewage treatment facilities. United Nations peace keepers from Nepal came to the country to help with the disaster. An epidemic of cholera broke out 10 months after the earthquake affecting hundreds of thousands of people. Cholera was not endemic to Haiti. Cholera is caused by the gram negative bacteria *Vibrio cholerae* and causes severe dehydration due to diarrhoea.



- (i) Define the terms *endemic* and *epidemic*. [2]

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- (ii) Explain how cholera spread in Haiti. [2]

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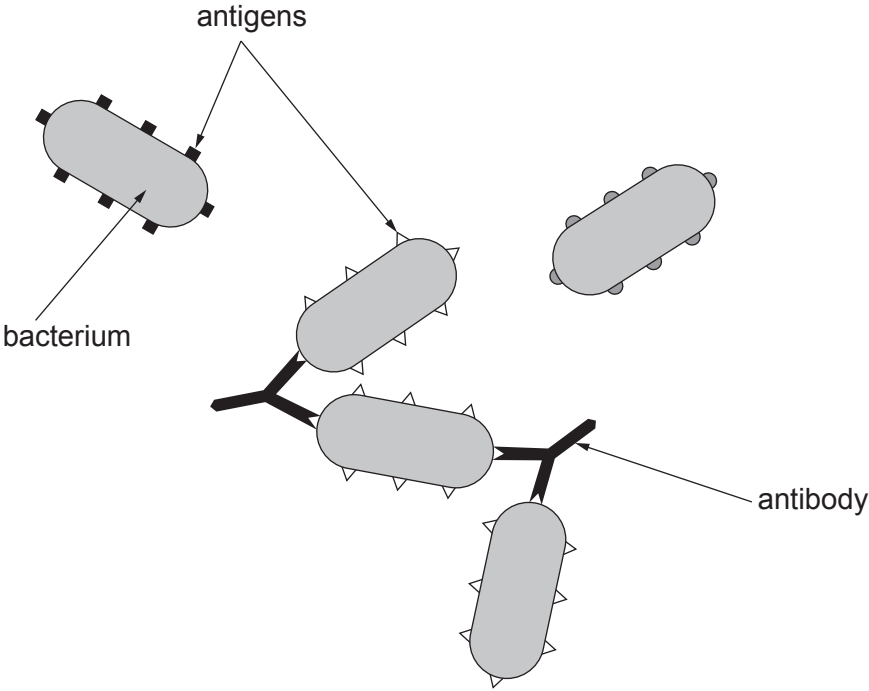
- (iii) Conclude what could have been the original source of the cholera epidemic, explain your answer. [1]

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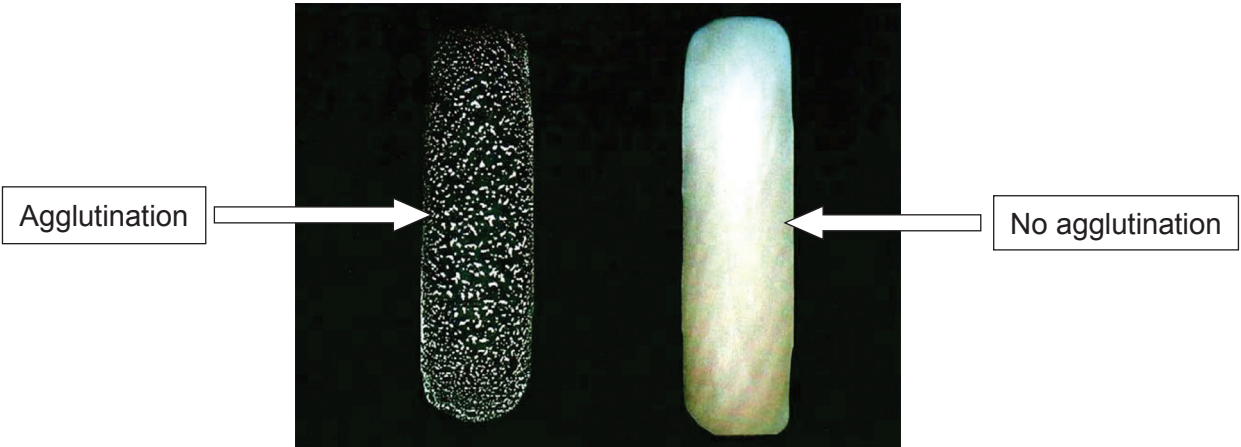
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- (b) Cholera is caused by the production of cholera toxin by 2 strains of *V. cholerae*, O1 and O139. An agglutination test can be carried out to test for the presence of O antigens on the bacterial surface. Agglutination is when clumping takes place between antibodies and antigens. A different agglutination test can be carried out to test for each of the strains.



Samples were taken from patients for testing to confirm the strain of bacteria and determine the source of the outbreak.



Examiner
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(i) Explain why an agglutination test would be able to distinguish between the two strains of cholera bacteria. [3]

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(ii) How would the scientists be able to confirm the source of the outbreak? [2]

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(iii) Suggest why some patients with severe symptoms are given antibiotic tablets, but antibiotics alone are not a cure for the disease. [3]

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(iv) Treatment for severe dehydration caused by Cholera is by intravenous rehydration. Patients are given 200 cm³ of fluid per kg of body mass in a 24 hour period. Calculate the volume of liquid per hour to be given to a patient weighing 70 kg. [2]

Volume of liquid per hour = cm³



Examiner
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(c) Oral cholera vaccines can be used to prevent the disease if there is a risk of an epidemic. Two doses of the vaccine are required. Shancol is a vaccine which has been used successfully in areas of endemic cholera but was not used during the Haiti epidemic as it had not then been authorised for use by the United Nations/World Health Organisation (UN/WHO).

(i) Explain why the oral vaccine must be administered in two doses. [2]

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(ii) State **one** reason why a higher concentration of the vaccine must be used when given orally. [1]

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(iii) What considerations would need to be made by the UN/WHO before allowing the use of a vaccine in Haiti? [2]

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